COMPUTATIONAL PROBLEM SOLVING 2

Project 1

Report

TEAM MEMBERS

Elie Kalouche • Zayne Howley • Kyle Goucher

1. Software Design and Architecture

1.1 Main Program and User Interface

Main.cpp consists of three main components, these being initialization/exit procedures, the main application loop in main(), and the search sub-menu in DisplaySearchMenu(). Upon initialization, a Database object is dynamically allocated with a pointer storing its address to access Database methods. The vehicle inventory is then loaded from an external “Inventory.txt” file. Before exiting main(), the Database object is deallocated using a delete statement. The main function uses a do-while loop to keep the program running until the user requests to exit. It is responsible for displaying the main menu and the program’s current date. A switch-case statement is used to process user commands. DisplaySearchMenu() is a separate function that handles the logic of the searching sub-menu, using a switch-case statement of its own to help the user search by various categories. Main is focused purely on the user interface, delegating the logic of the car reservation system to Database and accessing its methods when required.

The main menu provides the user with seven options, these being:

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles with Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit

The search menu provides the user with six more options, these being:

1. Search by Make
2. Search by Model
3. Search by Year
4. Search by Color
5. Search by Propulsion Type
6. Return to Main Menu

The information was presented in this way to provide the user with a clean and easily readable interface that meets all the requirements for the car reservation system. The main menu and search menu were separated to prevent the main menu from becoming too cluttered.

1.2 Module Class Hierarchy

The Vehicle, Gas Vehicle, and Electric Vehicle classes store vehicle information using inheritance. The Vehicle class is the base class and contains properties such as make, model, color, propulsion type, and year, using setter and getter functions to update and retrieve these

values. Its constructor initializes the string fields to empty and the year stored as an integer to zero. It also contains a virtual destructor to support the deletion of derived objects through Vehicle pointers. The Electric and Gas Vehicle classes though inherit the shared properties and methods from the Vehicle class but also have their own properties, those being capacity and efficiency stored as a double and its constructor initializes its values to zero. Both classes do differ slightly, though the Gas Vehicle class stores its fuel capacity in gallons and its efficiency in miles per gallon, while the Electric Vehicle class stores the battery capacity in kilowatt hours and efficiency in miles per kilowatt hour but both classes do override the GetPropulsionInfo() function from the Vehicle class to calculate the driving range by multiplying the capacity by the efficiency and returning a summary of the propulsion type,range and efficiency using an output string that can be called on by the database or any other class later on.

1.3 Database Management

The database class is responsible for the process of storing vehicle information and managing the reservations. It contains a 2D array called *VehicleInfo* used to store details about the different vehicles. The rows are the different vehicles, and the columns represent the different details (which contain the make, model, year, color, propulsion type, range, and efficiency). Pointers are also used to point to the actual gas and electric vehicle objects. This allows for a better organization of data so the program can function properly.

Reservations are managed using a separate array that stores the vehicle number along with the day reserved. Each row represents a reservation and allows enough space for the number of vehicles to have up to the max number of reservations, while columns show the vehicle number and the day that is reserved. For example, if there are 25 vehicles in the list, there is enough space for 75 reservations to be made.

The simulated date is also managed to act as if the program progresses to the next day and comes across a day there is a reservation. The *AdvanceDate* method moves the simulated date forward and removes any reservations that have expired.

Other methods used to complete the task of data management include *ReadInventory*, which reads in vehicle data from the inventory file and creates corresponding objects. The *Search* functions allow the user to find vehicles based on the vehicle features (make, model, year, color, propulsion type, range, and efficiency). *ReserveVehicle* records a reservation, while *IsVehicleAvailable* checks if a vehicle is allowed to be reserved.

The database class was created so that the vehicle and reservation logic could be handled in a singular location and not be contained within the *Main.cpp*, which was how it was done in Lab 5. This theoretically should make the code easier to maintain and modify if it needs to be done.

2. Testing Methodology

The code was tested manually through the user interface by entering different inputs and observing the resulting output. Many scenarios were created to check the vehicle searches, reservations, vehicle display, and data advancement. The observed results were compared with the expected outputs to verify that the code was working and behaved correctly.

3.1 User Interface Testing

Figures 1–2 show the initial main menu and vehicle-search menu.

```
============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): |
```

Figure 1: Initial Screen

```
============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): 1

============== SEARCH VEHICLES ==============
1. Search by Make
2. Search by Model
3. Search by Year
4. Search by Color
5. Search by Propulsion Type (ICE/EV)
6. Return to Main Menu
Enter Choice (1-6):
```

Figure 2: Initial Search Screen

3.2 Vehicle Search Testing

Figures 3–12 cover matching and nonmatching searches across the five supported search fields.

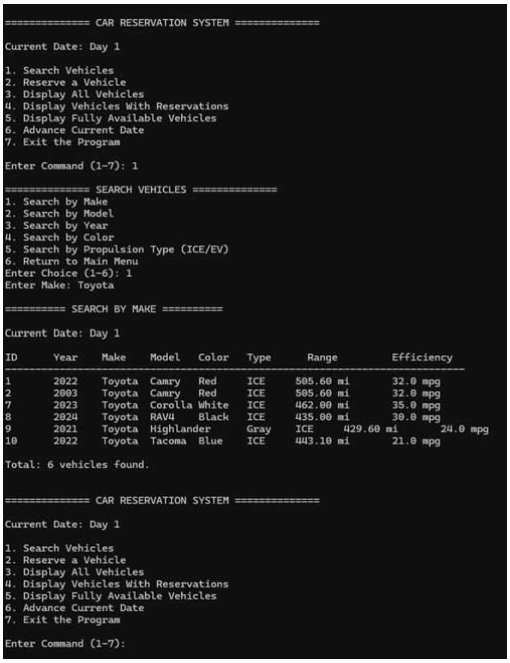


Figure 3: Searching by Make Successfully

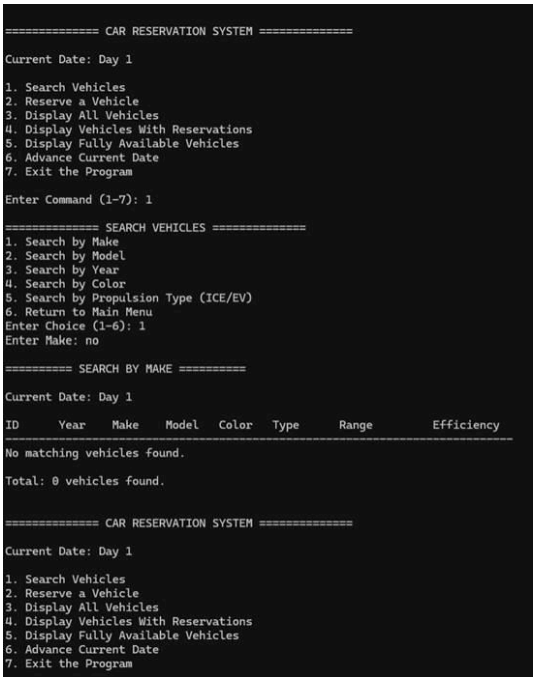


Figure 4: Searching by Make Unsuccessfully

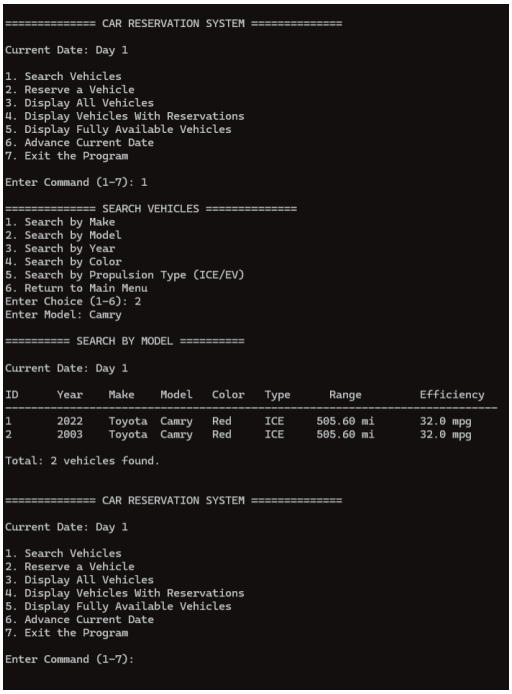


Figure 5: Searching by Model Successfully

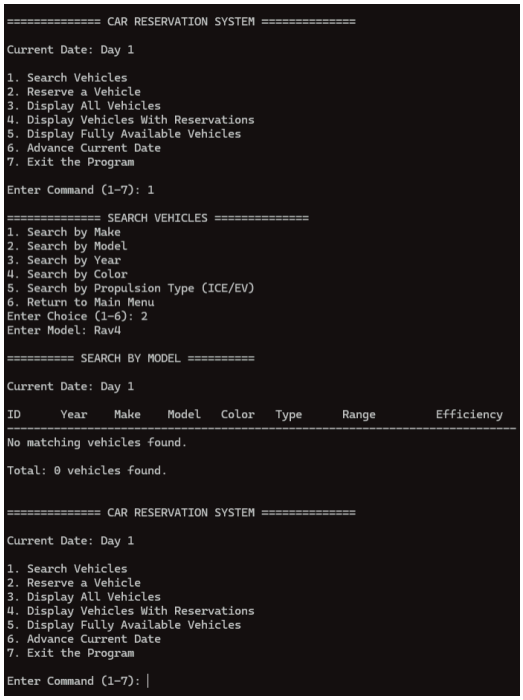


Figure 6: Searching by Model Unsuccessfully

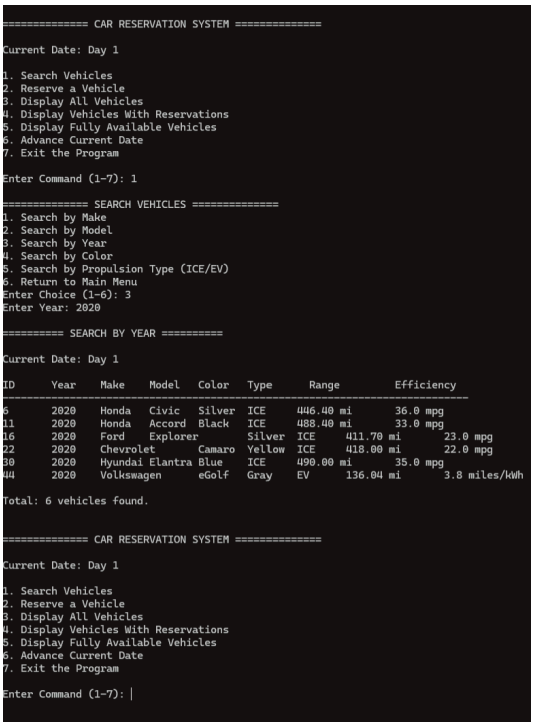


Figure 7: Searching by Year Successfully

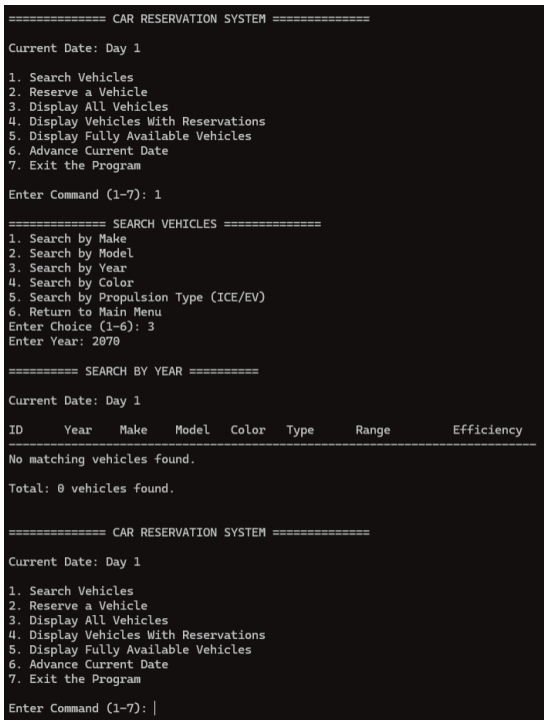


Figure 8: Searching by Year Unsuccessfully

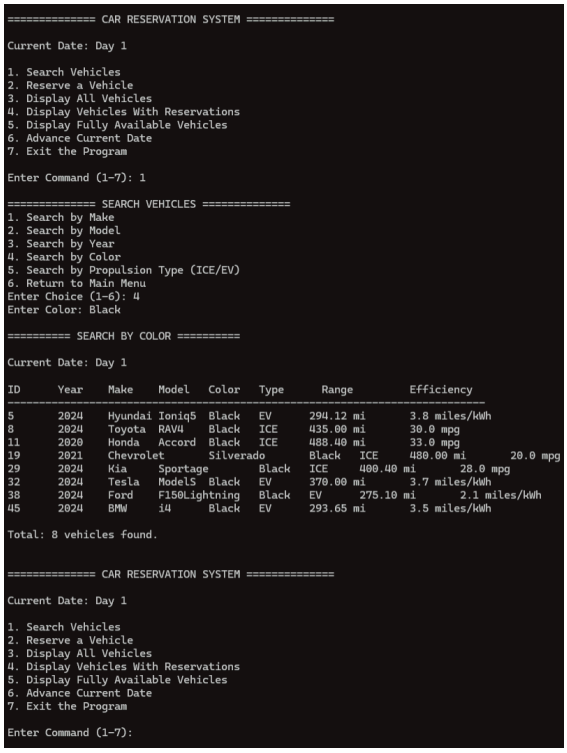


Figure 9: Searching by Color Successfully



Figure 10: Searching by Color Unsuccessfully

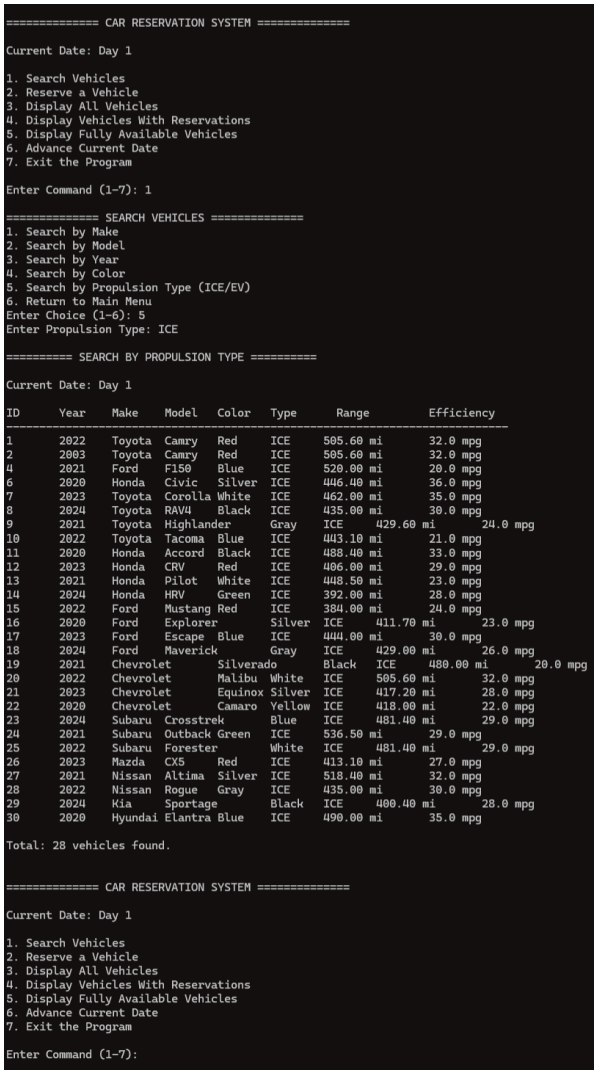


Figure 11: Searching by Propulsion Type Successfully

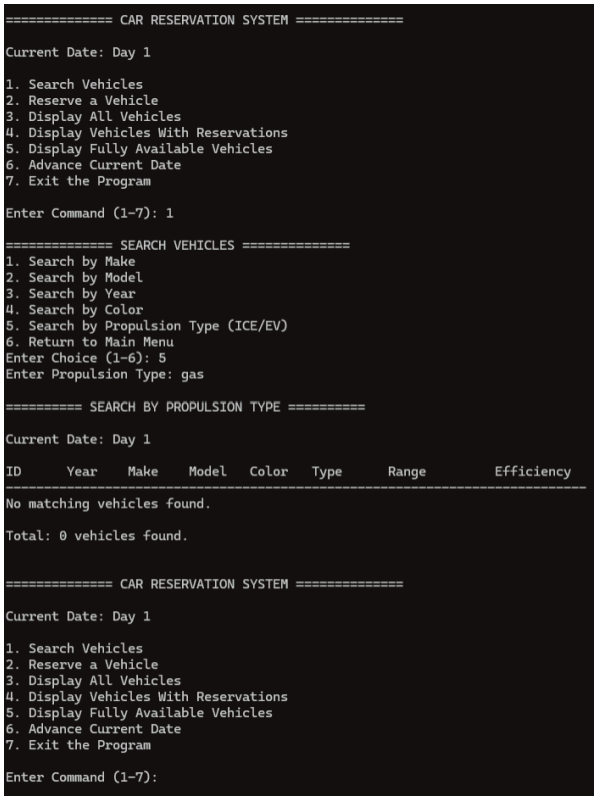


Figure 12: Searching by Propulsion Type Unsuccessfully

3.3 Vehicle Display Testing

These tests cover display of the entire inventory, vehicles with reservations, and vehicles that are fully available.

```
============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): 3

========== ALL VEHICLES ==========

Current Date: Day 1
Total Vehicles: 50

ID      Year    Make    Model   Color   Type      Range         Efficiency
-------------------------------------------------------------------------
1       2022    Toyota  Camry   Red     ICE     505.60 mi       32.0 mpg
2       2003    Toyota  Camry   Red     ICE     505.60 mi       32.0 mpg
3       2023    Tesla   Model3  White   EV      307.50 mi       4.1 miles/kWh
4       2021    Ford    F150    Blue    ICE     520.00 mi       20.0 mpg
5       2024    Hyundai Ioniq5  Black   EV      294.12 mi       3.8 miles/kWh
6       2020    Honda   Civic   Silver  ICE     446.40 mi       36.0 mpg
7       2023    Toyota  Corolla White   ICE     462.00 mi       35.0 mpg
8       2024    Toyota  RAV4    Black   ICE     435.00 mi       30.0 mpg
9       2021    Toyota  Highlander      Gray    ICE     429.60 mi       24.0 mpg
10      2022    Toyota  Tacoma  Blue    ICE     443.10 mi       21.0 mpg
11      2020    Honda   Accord  Black   ICE     488.40 mi       33.0 mpg
12      2023    Honda   CRV     Red     ICE     406.00 mi       29.0 mpg
13      2021    Honda   Pilot   White   ICE     448.50 mi       23.0 mpg
14      2024    Honda   HRV     Green   ICE     392.00 mi       28.0 mpg
15      2022    Ford    Mustang Red     ICE     384.00 mi       24.0 mpg
16      2020    Ford    Explorer        Silver  ICE     411.70 mi       23.0 mpg
17      2023    Ford    Escape  Blue    ICE     444.00 mi       30.0 mpg
18      2024    Ford    Maverick        Gray    ICE     429.00 mi       26.0 mpg
19      2021    Chevrolet       Silverado       Black   ICE     480.00 mi       20.0 mpg
20      2022    Chevrolet       Malibu  White   ICE     505.60 mi       32.0 mpg
21      2023    Chevrolet       Equinox Silver  ICE     417.20 mi       28.0 mpg
22      2020    Chevrolet       Camaro  Yellow  ICE     418.00 mi       22.0 mpg
23      2024    Subaru  Crosstrek       Blue    ICE     481.40 mi       29.0 mpg
24      2021    Subaru  Outback Green   ICE     536.50 mi       29.0 mpg
25      2022    Subaru  Forester        White   ICE     481.40 mi       29.0 mpg
26      2023    Mazda   CX5     Red     ICE     413.10 mi       27.0 mpg
27      2021    Nissan  Altima  Silver  ICE     518.40 mi       32.0 mpg
28      2022    Nissan  Rogue   Gray    ICE     435.00 mi       30.0 mpg
29      2024    Kia     Sportage        Black   ICE     400.40 mi       28.0 mpg
30      2020    Hyundai Elantra Blue    ICE     490.00 mi       35.0 mpg
31      2022    Tesla   ModelY  Blue    EV      285.00 mi       3.8 miles/kWh
32      2024    Tesla   ModelS  Black   EV      370.00 mi       3.7 miles/kWh
33      2021    Tesla   ModelX  White   EV      320.00 mi       3.2 miles/kWh
34      2023    Chevrolet       Bolt    Silver  EV      260.00 mi       4.0 miles/kWh
35      2022    Nissan  Leaf    Red     EV      152.00 mi       3.8 miles/kWh
36      2024    Nissan  Ariya   Gray    EV      269.70 mi       3.1 miles/kWh
37      2023    Ford    MustangMachE    Blue    EV      273.00 mi       3.0 miles/kWh
38      2024    Ford    F150Lightning   Black   EV      275.10 mi       2.1 miles/kWh
39      2022    Hyundai KonaElectric    White   EV      256.00 mi       4.0 miles/kWh
40      2024    Hyundai Ioniq6  Silver  EV      325.08 mi       4.2 miles/kWh
41      2023    Kia     EV6     Red     EV      286.38 mi       3.7 miles/kWh
42      2024    Kia     NiroEV  Blue    EV      252.72 mi       3.9 miles/kWh
43      2023    Volkswagen      ID4     White   EV      262.40 mi       3.2 miles/kWh
44      2020    Volkswagen      eGolf   Gray    EV      136.04 mi       3.8 miles/kWh
45      2024    BMW     i4      Black   EV      293.65 mi       3.5 miles/kWh
46      2023    BMW     iX      Blue    EV      323.35 mi       2.9 miles/kWh
47      2024    Cadillac        Lyriq   Red     EV      306.00 mi       3.0 miles/kWh
48      2023    Rivian  R1T     Green   EV      310.50 mi       2.3 miles/kWh
49      2024    Rivian  R1S     White   EV      297.00 mi       2.2 miles/kWh
50      2022    Porsche Taycan  Silver  EV      261.52 mi       2.8 miles/kWh

Total: 50 vehicles found.


============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7):
```

Figure 13: Display All Vehicles Successfully

```
============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): 4

========== VEHICLES WITH RESERVATIONS ==========

Current Date: Day 1

ID      Year    Make    Model   Color   Type      Range Efficiency              Reserved Days
-------------------------------------------------------------------------------------------
4       2021    Ford    F150    Blue    ICE     520.00 mi       20.0 mpg      2
40      2024    Hyundai Ioniq6  Silver  EV      325.08 mi       4.2 miles/kWh 3
49      2024    Rivian  R1S     White   EV      297.00 mi       2.2 miles/kWh 2

Total Reserved Vehicles: 3
Total Reservations: 3


============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7):
```

Figure 14: Display All Vehicles with Reservations Successfully

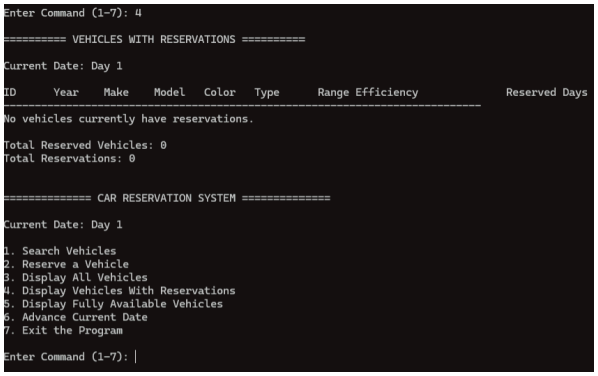


Figure 15: Display All Vehicles with Reservations Unsuccessfully

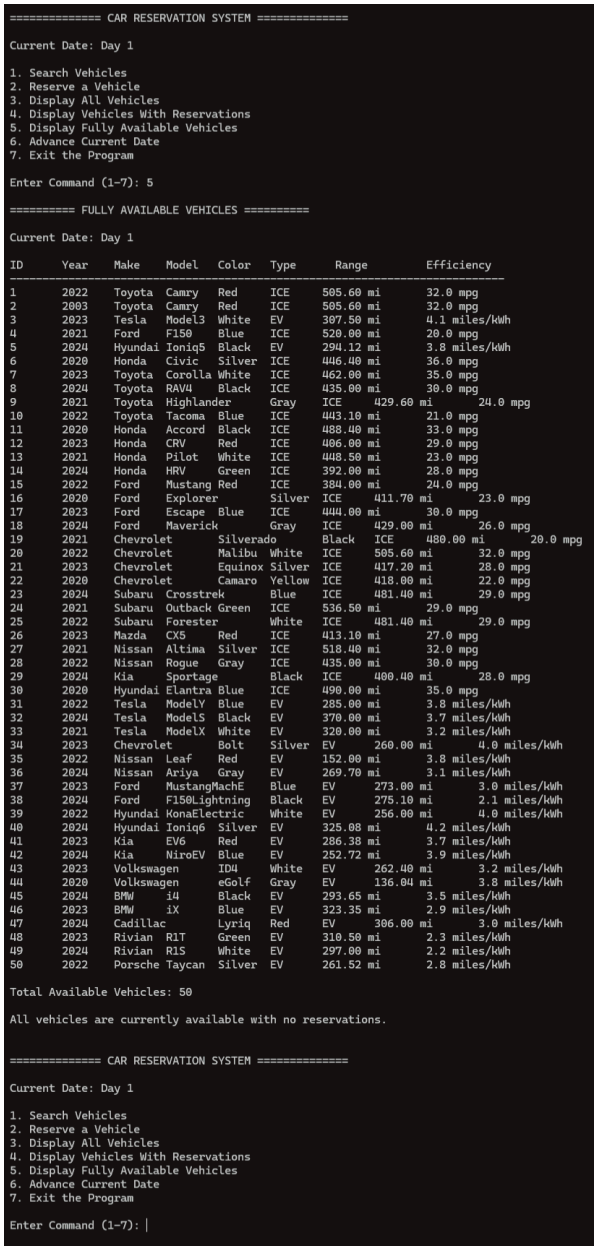


Figure 16: Display All Fully Available Vehicles Successfully

3.4 Reservation Testing

These cases document reservation behavior and handling of the maximum number of reserved vehicles.

```
============== CAR RESERVATION SYSTEM ==============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): 2

Enter Vehicle ID: 1
Enter Reservation Day: 2
Success: Vehicle #1 reserved for Day 2.
```

Figure 17: Reserve a Vehicle Successfully

```
============= CAR RESERVATION SYSTEM =============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): 2

Enter Vehicle ID: 1
Enter Reservation Day: 1
Error: Reservation failed. Number of reservation days exceeded (more than 3 days)
```

Figure 18: Reserve a Vehicle Unsuccessfully

```
============= CAR RESERVATION SYSTEM =============

Current Date: Day 1

1. Search Vehicles
2. Reserve a Vehicle
3. Display All Vehicles
4. Display Vehicles With Reservations
5. Display Fully Available Vehicles
6. Advance Current Date
7. Exit the Program

Enter Command (1-7): 2

Enter Vehicle ID: 2
Enter Reservation Day: 5
Error: Reservation failed. Number of reservation days exceeded (more than 3 days)
```

Figure 19: More Than 3 Reservations Per Vehicle Error

3.5 Date Management Testing

The following figures address changing the current date.

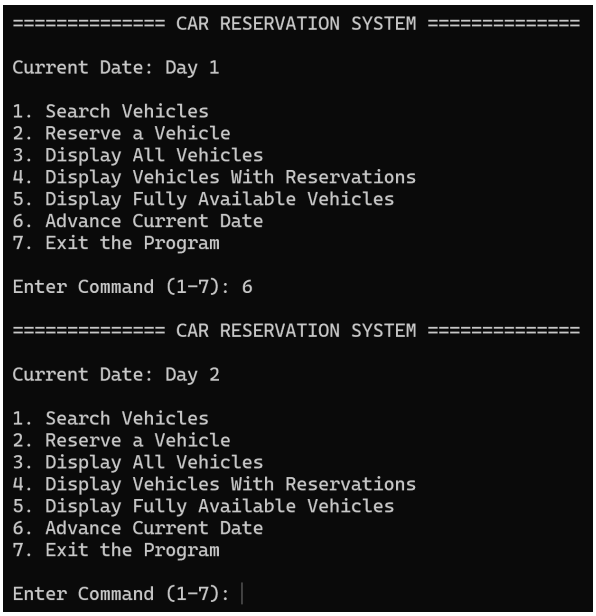


Figure 20: Advance Current Date Successfully

3.6 Input and File Error Handling

These cases cover oversized inventory files, files that cannot be found, and invalid file formats.

```
2023 Ford MustangMachE EV Blue 91.0 3.0
2024 Ford F150Lightning EV Black 131.0 2.1
2022 Hyundai KonaElectric EV White 64.0 4.0
2024 Hyundai Ioniq6 EV Silver 77.4 4.2
2023 Kia EV6 EV Red 77.4 3.7
2024 Kia NiroEV EV Blue 64.8 3.9
2023 Volkswagen ID4 EV White 82.0 3.2
2020 Volkswagen eGolf EV Gray 35.8 3.8
2024 BMW i4 EV Black 83.9 3.5
2023 BMW iX EV Blue 111.5 2.9
2024 Cadillac Lyriq EV Red 102.0 3.0
2023 Rivian R1T EV Green 135.0 2.3
2024 Rivian R1S EV White 135.0 2.2
2022 Porsche Taycan EV Silver 93.4 2.8
2023 Porsche Taycan EV Silver 93.4 2.8
```

```
Error! Inventory contains more than 50 vehicles.
Failed to initialize database inventory. Exiting program.
```

Figure 21: Too Many Vehicles in File Error

```
Error! Inventory file could not be opened.
Failed to initialize database inventory. Exiting program.
```

Figure 22: File Does Not Exist Error

```
Error! Inventory file could not be opened.
Failed to initialize database inventory. Exiting program.

D:\Visual Studio Code Project Libary\EBHComp2\Reservation_System_V1\ReservationSystem\x64\Debug\ReservationSystem.exe (process
 32196) exited with code -1 (0xffffffff).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close the console whe
n debugging stops.
Press any key to close this window . . .
```

Figure 23: Wrongly Formatted File Error